

EE-482: Network Security and Cryptography

LECTURE NO 3: *Conventional Cryptography*

Engr. Tehseen Ahsan

**Assistant Professor, Computer Engineering Department
HITEC University Taxila Cantt, Pakistan**

Organization of Lecture

1. Cryptography
2. Conventional Encryption Principles
3. Model of Conventional Encryption
4. Security of Encryption
5. Classification of Cryptographic Systems
 - a. Substitution Ciphers
 - b. Transposition Techniques
6. Steganography

1. Cryptography

* CRYPTOGRAPHY

Cryptography is the art of achieving security by encoding messages to make them non-readable

Why Cryptography?

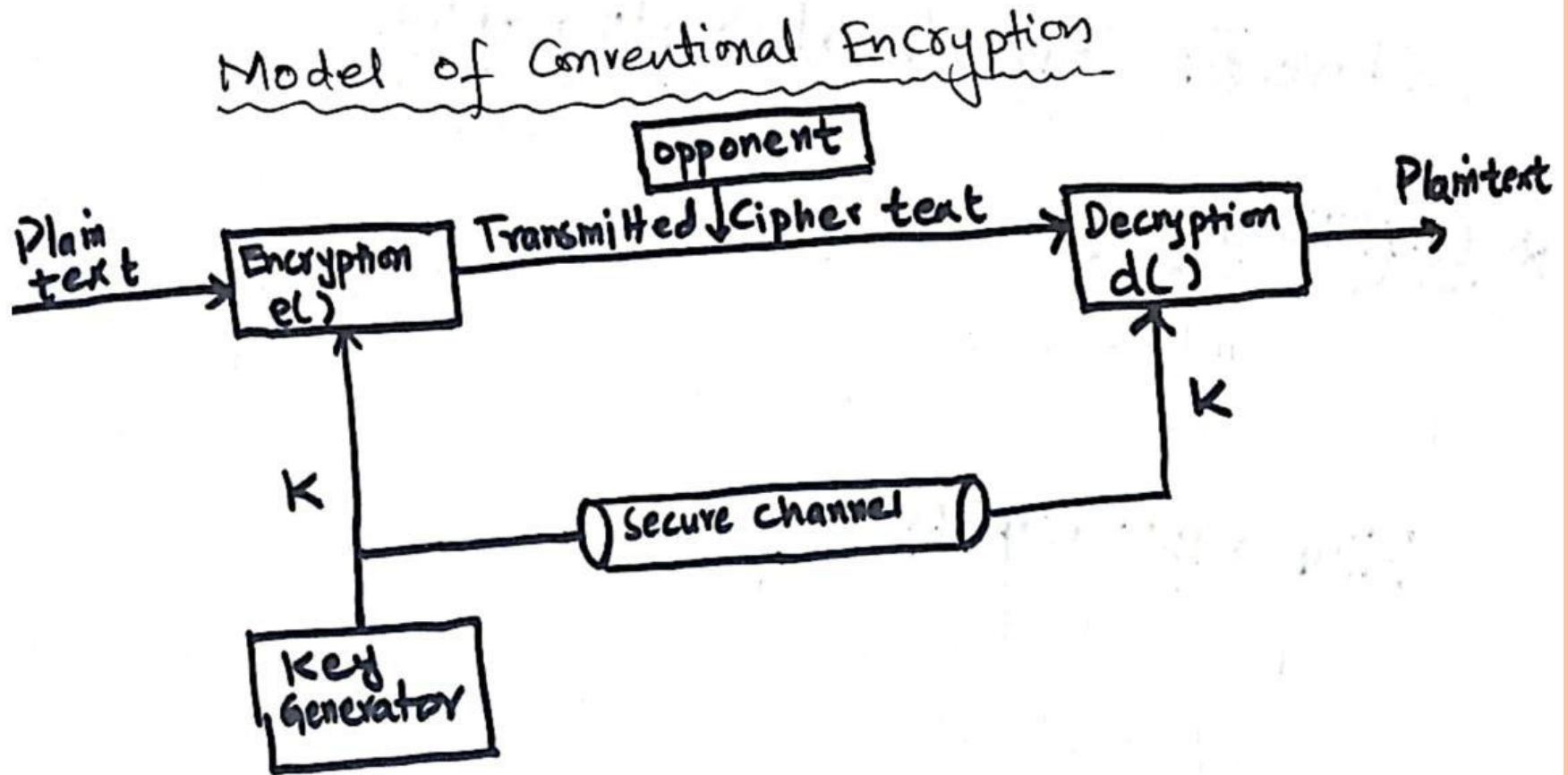
- 1) Confidentiality: Keep information secret
- 2) Authentication: Receiver can verify who sender was
- 3) Integrity: detect modified messages.
- 4) Non-repudiation: Sender cannot falsely deny sending a message.
Receiver cannot deny receiving

2. Conventional Encryption Principles

* CONVENTIONAL ENCRYPTION PRINCIPLES:

- 1) Plaintext: This is the original message or data.
- 2) Encryption algorithm: Performs various substitutions and transformations on the plain text
- 3) Secret key: It is also input to the algorithm. The exact substitution or transformation depends on the key
- 4) Cipher text: Scrambled/Encoded message produced as an output of encryption.
2 different keys will produce two different cipher texts
- 5) Decryption algorithm: The encryption algorithm runs in reverse & outputs original plaintext.

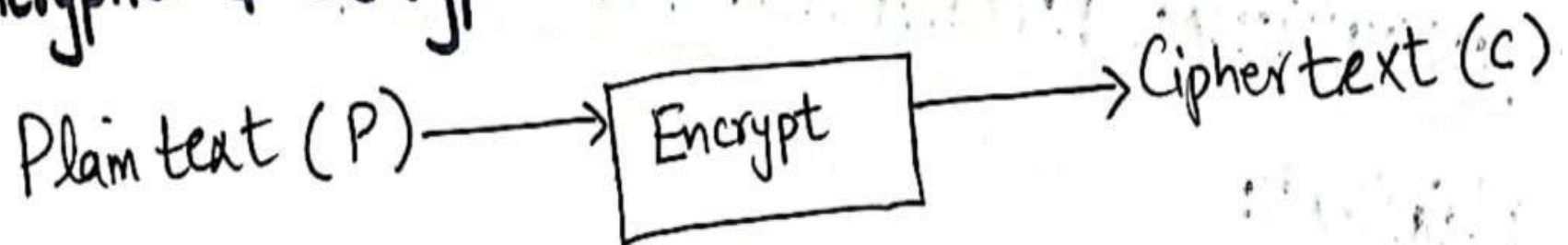
3. Model of Conventional Encryption (1/4)



* The above model is also known as:
Symmetrical key cryptosystem
or Single key cryptosystem
Or Shared Secret
Cryptosystem

3. Model of Conventional Encryption (2/4)

Encryption & Decryption:



$$C = e(P)$$



$$P = d(C)$$

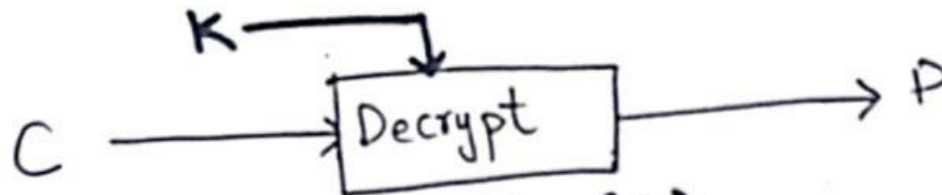
$$P = D(E(P))$$

3. Model of Conventional Encryption (3/4)

Encryption with a Secret key:



$$C = E_K(P)$$

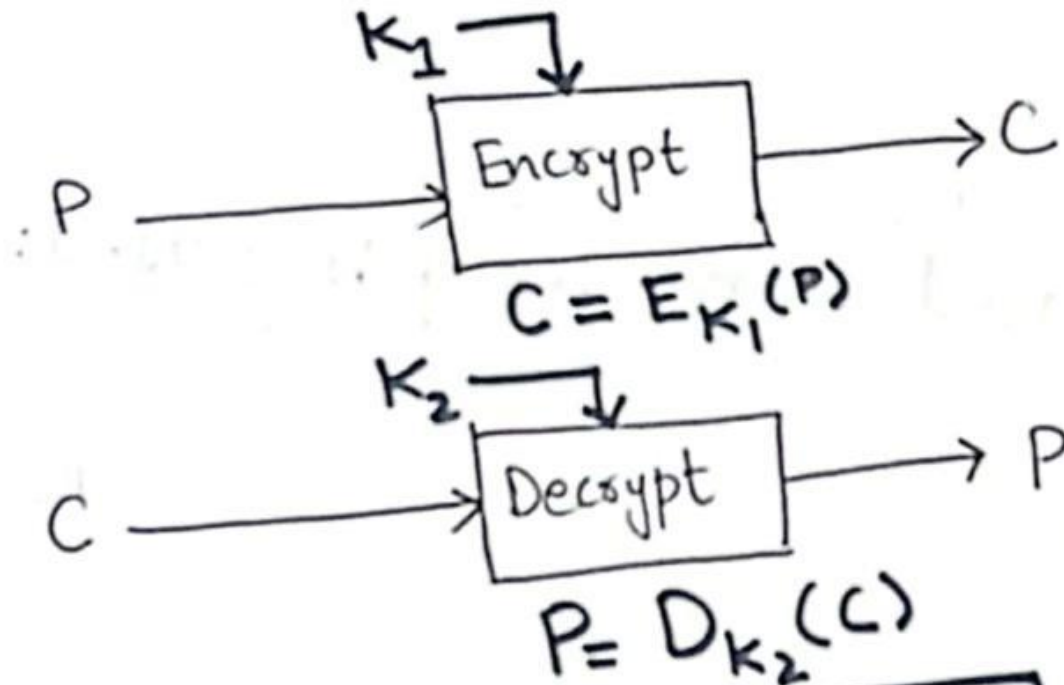


$$P = D_K(C)$$

$$P = D_K[E_K(P)]$$

3. Model of Conventional Encryption (4/4)

Encryption with 2 keys:



$$P = D_{K_2}[E_{K_1}(P)]$$

4. Security of Encryption

SECURITY OF ENCRYPTION:

- ① Encryption algorithm must be powerful enough. Opponent should not be able to decrypt ciphertext or discover the key
- ② Security depends on the secrecy of key and not on the secrecy of algorithm.

5. Classification of Cryptographic Systems (1/23)

CLASSIFICATION OF CRYPTOGRAPHIC SYSTEMS

(a) Classification based on type of operation:

↳ Type of operation used for transforming plaintext to cipher text

Substitution (mapping)

Transposition (rearrangement)

⇒ Requirement: no. information be lost i.e., all operations be reversible

5. Classification of Cryptographic Systems (2/23)

(b) Classification based on number of keys used:

↳ Symmetric / Single key / secret key / conventional encryption:
(sender & receiver use the same key).

↳ Asymmetric / two-key / public key encryption:
(sender & receiver use different keys).

(c) Classification based on the way plaintext is processed:

↳ Block cipher: processes input as one block of elements at a time producing an output block for each input block.

↳ Stream cipher: processes input elements continuously producing one element at a time.

Block
ATTACK POSTPONED

Stream
ATTACK

5. Classification of Cryptographic Systems (3/23)

* Two primary ways in which we can code a plaintext message into cipher text.



NOTE: When both approaches used together "PRODUCT CIPHER"

SUBSTITUTION CIPHERS

The elements are systematically replaced by other elements (characters, numbers, symbols)

5. Classification of Cryptographic Systems (4/23)

1) Caesar Cipher:

↳ this scheme first proposed by Julius Caesar
↳ each alphabet in a message is replaced by an alphabet 3 places down the line.

A B C D E F G H I J K L M N O P Q



Plain Text: A T T A C K

POSTPONED
SRVWSRQHGI

Cipher Text: D W W D F N

For Encryption: $\text{Key} = +3$

For Decryption: $\text{Key} = -3$ (move back 3 places)

Disadvantage: very weak algorithm.

5. Classification of Cryptographic Systems (5/23)

2) Mono alphabetic Cipher:

↳ Each character is replaced by any of the remaining character

A $\longrightarrow$ B; ... Z

B $\longrightarrow$ A; C ... Z

C $\longrightarrow$ A; B; D ... Z

* **Cryptanalysis:** The process of attempting to discover plain text or key.

Cryptanalyst: A person who attempts to break a cipher text

5. Classification of Cryptographic Systems (6/23)

* **Brute Force Attack:** Try all possibilities to decrypt the cipher text.

Caesar cipher : 25 possibilities.

Monoalphabetic cipher: can have any combination.

$$25! = 25 \times 24 \times 23, \dots$$

5. Classification of Cryptographic Systems (7/23)

Monoalphabetic using a key word:

↳ Use a letter only the first time it appears in a key word.

Alphabets:

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
A	N	D	R	E	N	C	I	P	H	B	F	G	J	K	L	M	O	Q	S
U	V	W	X	Y	Z														
T	U	V	X	Y	Z														

Key: ANDREW CIPHER

Plaintext: GOOD MORNING } Encryption.

Cipher text:

CKKR G KOJPJC

CipherText: C K K R G K O J P J C

Plain Text: GOOD MORNING.

Cryptanalyst can make use of relative frequency of characters.

5. Classification of Cryptographic Systems (8/23)

3) Vigenère tableau

↳ Table is given in next slide

- * It is also called as Polyalphabetic substitution.
- * Pick a key (e.g. the key will be "MEC")
- * Let's take plain text as follows:
Plain Text: WE NEED MORE SUPPLIES FAST
- * Write key word across the top of the text you want to encrypt, repeating it as many times as necessary.

5. Classification of Cryptographic Systems (9/23)

VIGENERE TABLE

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

5. Classification of Cryptographic Systems (10/23)

Key: M E C M E C M E C M E C M E C M
 Plaintext: W E N E E D M O R E S U P P L I E S F A S T
 Cipher text: I I P Q I F Y S T Q W W B T N U I U R E U F

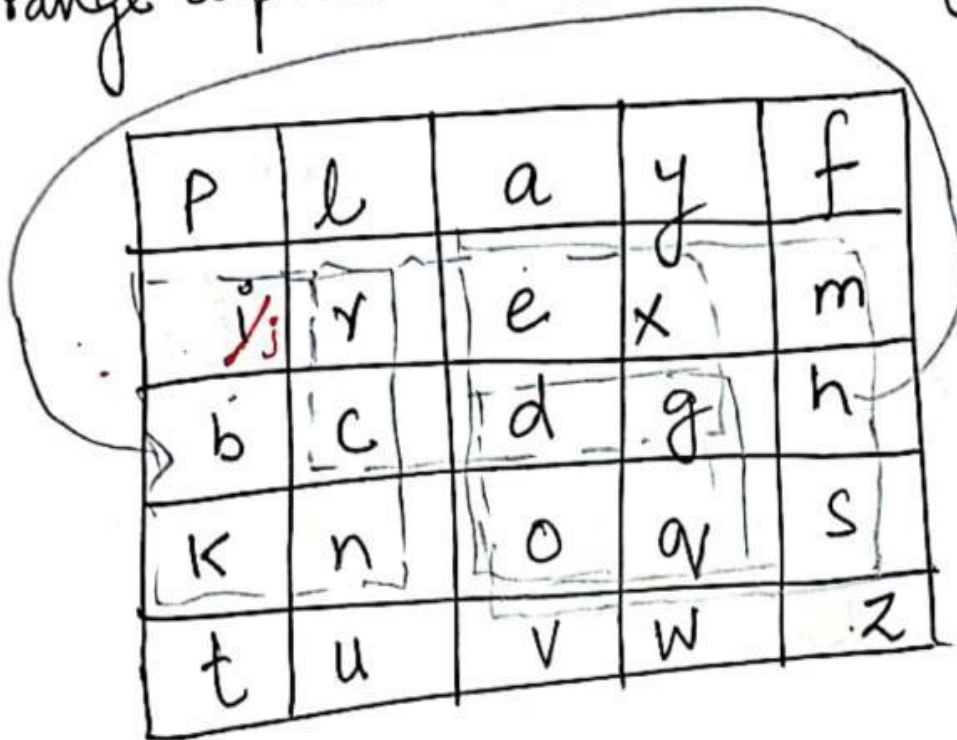
- * For each plaintext alphabet, look at the alphabet of the key above plaintext (e.g. if it was 'M', then you would go to the row that starts with an 'M'), and find that row in the Vignere table.
- * Then find the column of plaintext letter (for example, 'W')
- * Finally, trace down that column until you reach the row that you found before and write down the letter in the cell where they intersect.

5. Classification of Cryptographic Systems (11/23)

4) Play Fair Cipher:

↳ also called Polygraph Cipher

↳ arrange alphabets into a 5X5 grid (Total 25 entries)



P	l	a	y	f
i / j	r	e	x	m
b	c	d	g	h
k	n	o	q	s
t	u	v	w	z

5. Classification of Cryptographic Systems (12/23)

↳ the key is filled/inserted in table above

↳ Assume key as: Playfair example

↳ Procedure for filling the matrix is:

1. Top ; left $\rightarrow$ right

2. Repeated letters/alphabets are ignored.

3. Fill the empty field of above matrix by characters in alphabetical order with condition that both i/j occupies a single cell.

5. Classification of Cryptographic Systems (13/23)

Encryption:

Rules:

1. Divide the plain text into a pair of letters
↳ e.g. Welcome
we|lc|om|ex

2. Pair contains repeated letters, replace the repeated letter by 'x'

↳ Hello

He|ll|o

Helxlo ⇒ Between the repeated letters, include an additional letter 'x'.

He|lx|lo

3. If two letters are in same row.
↳ Replace them with immediate right letter.

5. Classification of Cryptographic Systems (14/23)

4. If two letters are in same column
↳ Replace them with immediate below letter
5. If two letters are not in same row or in same column.
↳ Draw a rectangle / square as corners of plaintext & remaining two corners on same row becomes cipher Text.

5. Classification of Cryptographic Systems (15/23)

Ex 1

(P.T) Plain Text: cd
(C.T) Cipher Text: dg } Rule 3

Ex 4

P.T: dv
C.T: oa } Rule 4

Ex 2

P.T: ch
C.T: db } Rule 3

Ex 5

P.T: cn
C.T: gr } Rule 5

Ex 3

P.T: ed
C.T: do } Rule 4

5. Classification of Cryptographic Systems (16/23)

Ex 6

Good morning, Assume key = Playfair example

Solⁿ

Good morning in
day vo es cu rk

PT = Plain Text = P.T
CT = Cipher Text = C.T

P.T = Good morning
C.T = dayvoescurk

5. Classification of Cryptographic Systems (17/23)

5) Free mason Cipher:

↳ Uses special symbols as replacement for alphabets

A	B	C
D	E	F
G	H	I

J
K L
M

N	O	P
Q	R	S
T	U	V

W
X Y
Z

P.T: UNIVERSITY
C.T: ▯ ▯ ▯ ▯ ▯ ▯ ▯ ▯ ▯ ▯

5. Classification of Cryptographic Systems (18/23)

6) Hill Cipher:

↳ makes use of system of linear equations.

↳ one-to-one correspondance b/w alphabets and numbers.

$$[A, B, C, \dots, Z] \mapsto [0, 1, 2, \dots, 25]$$

↳ Key is user-defined & remains fixed for encryption & decryption.

$$\begin{bmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{bmatrix} \begin{bmatrix} P_1 \\ P_2 \\ P_3 \end{bmatrix} = \begin{bmatrix} C_1 \\ C_2 \\ C_3 \end{bmatrix}$$

Key Plaintext Cipher Text

5. Classification of Cryptographic Systems (19/23)

Example

Plain Text: x y z
(23) (25)
(24)

Let's suppose user defined the key as matrix which is given below. Compute the Cipher Text.

$$\begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 1 & 1 & 2 \end{bmatrix} \times \begin{bmatrix} 23 \\ 24 \\ 25 \end{bmatrix} = \begin{bmatrix} 146 \\ 143 \\ 97 \end{bmatrix}$$

Encryption: $KP = C$

Decryption: $P = K^{-1}C$

$$\text{Where } K^{-1} = \frac{\text{adj } K}{|K|} = \frac{1}{|K|} \cdot \text{Adj } K.$$

Note: Key matrix should be invertible.

5. Classification of Cryptographic Systems (20/23)

TRANSPOSITION TECHNIQUES

↳ In transposition cipher the letters of the original message remain the same but their positions are scrambled in some systematic way

1) Rail Fence Technique:

write plain text as sequence of diagonals
and read it row by row

Plaintext: Meet me after the party.

M E E T M E A F T E R T H E P A R T Y X X

Length = 7

Depth = 3

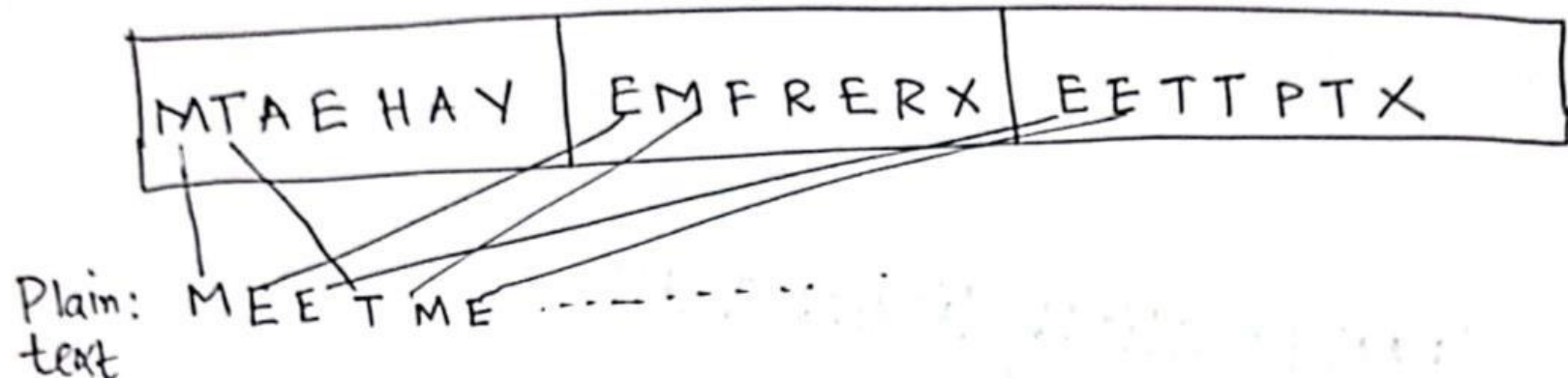
5. Classification of Cryptographic Systems (21/23)

Cipher Text: M T A E H A Y E M F R E R X E E T T P T X

Filler characters

$$\text{Decryption} = \frac{\text{Length of Cipher Text}}{\text{Depth}} = \frac{21}{3} = 7$$

* We will divide the cipher into 3 (depth) blocks, each consisting of 7 Letters



5. Classification of Cryptographic Systems (22/23)

2) Columnar Transposition:

* Simple columnar cipher:

P.T: Launch missile now

L A U N C

H M I S S

I L E N O

W X X X X

{ Specify no. of columns (length)
& rows, in this example rows=4, columns=5
(depth)

C.T: LhIwamlxuIeXnsnxcsoX

5. Classification of Cryptographic Systems (23/23)

* Columnar cipher using key

↳ key is user defined.

P.T: Attack postponed till two AM. { columns = length = 6
Rows = Depth = 4

1	2	3	4	5	6
A	T	T	A	C	K
P	O	S	T	P	O
N	E	D	T	I	L
L	T	W	O	A	M

Key: 4 3 5 2 1 6

C.T: ATTO TSDW CPIA TOET APNL KOLM
4 3 5 2 1 6

To decrypt need key, no. of rows, columns.

Decryption = Home Work.

6. Steganography (1/3)

* Stegano^agraphy:

C.T: $\rightarrow$ conceal/hide the message

Dear George

Greetings to all at Oxford. Many thanks for your letter and for the summer examination package

All entry forms and fees forms should be ready for final dispatch to the syndicate by Friday

20th or at least by the 21st.

There is, however still room for improvement. Just give us two or three years and we will really show you. Please don't let the proposals destroy your basic O and A patterns. Certainly this sort of change, if implemented immediately would bring disaster.

6. Steganography (2/3)

P.T: your package ready Friday 21st room three.
Please destroy this immediately disaster.

Steganography Implementation:

- ↳ Character marking
 - ↳ selected letters in text are written in different font/style
 - ↳ visible at an angle to light
- ↳ invisible ink.
- ↳ pin punctures.
- ↳ hide message in images

6. Steganography (3/3)

Limitations:

- 1) Requires a lot of overhead to hide relatively few bits of information.
 - 2) Once system is discovered, steganography becomes useless.
- ⇒ Steganography used by parties who don't want their identity revealed.
- ⇒ In cryptography the sender & receiver are known.

References: Textbooks

1. *“Cryptography and Network Security”, by Behrouz A. Forouzan, Debdeep Mukhopadhyay, 2nd Edition, McGraw Hill Education (India) Private Limited.*
2. *“Cryptography and Network Security Principles and Practice”, by William Stallings, 6th Edition, Pearson.*
3. *“Computer Security Principles and Practice”, by William Stallings, Lawrie Brown, 4th Edition, Pearson.*

Annexure:

Finding the Inverse of a Two-By-Two Matrix

$$A = \begin{bmatrix} 4 & 3 \\ 3 & 2 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} -2 & 3 \\ 3 & -4 \end{bmatrix}$$

$$A A^{-1} = \begin{bmatrix} 4 & 3 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} -2 & 3 \\ 3 & -4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \checkmark$$

the **identity matrix** confirms
this inverse relationship